

Ref.No. – FES/QTN/2024-25/0804

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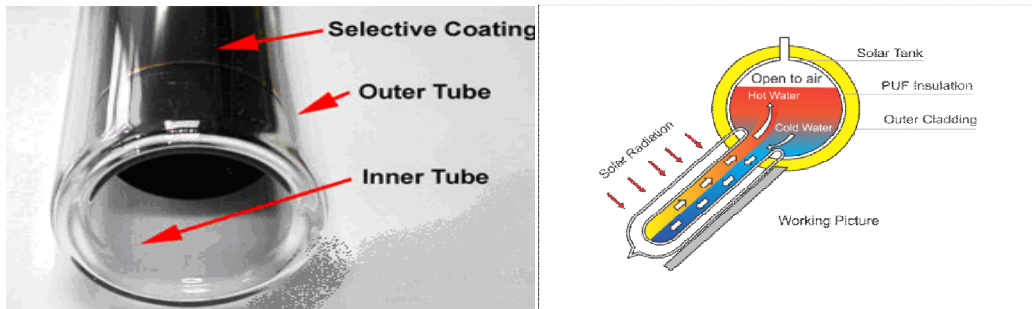
**Sub: Quotation for Solar Water Heating Systems – Reg.**

Dear Sir,

Thank you very much for the keen interest shown by you in our Sunlit Solar Water Heating Systems. We are pleased to enclose herewith the offer.

**SPECIFICATION FOR EVACUATED TUBES**

Evacuated tubes are the absorber of the solar water heater. They absorb solar energy converting it into heat for use in water heating. Evacuated tubes have already been used for years in Germany, Canada, China and the UK. There are several types of evacuated tubes in use in the solar industry. Sunlit collectors use the most common "twin-glass tube". This type of tube is chosen for its reliability, performance and low manufacturing cost.



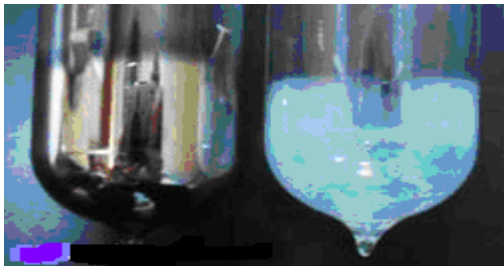
Each evacuated tube consists of two glass tubes made from extremely strong borosilicate glass. The outer tube is transparent allowing light rays to pass through with minimal reflection. The inner tube is coated with a special selective coating (CU-N/Al) which features excellent solar radiation absorption and minimal reflection properties. The top of the two tubes are fused together and the air contained in the space between the two layers of glass is pumped out while exposing the tube to high temperatures. This "evacuation" of the gasses forms a vacuum, which is an important factor in the performance of the evacuated tubes.

Why a vacuum? As you would know if you have used a glass lined thermos flask, a vacuum is an excellent insulator. This is important because once the evacuated tube absorbs the radiation from the sun and converts it to heat; we don't want to lose it!! The vacuum helps to achieve this. The insulation properties are so good that while the inside of the tube may be 150°C / 304°F, the outer tube is cold to touch. This means that evacuated tube water heaters can perform well even in cold weather when flat plate collectors perform poorly due to heat loss (during high Delta-T conditions)

In order to maintain the vacuum between the two glass layers, a barium getter is used (the same as in television tubes). During manufacture of the evacuated tube this getter is exposed to high temperatures which cause the bottom of the evacuated tube to be coated with a pure layer of barium. This barium layer actively absorbs any CO, CO<sub>2</sub>, N<sub>2</sub>, O<sub>2</sub>, H<sub>2</sub>O and H<sub>2</sub> out-gassed from the evacuated tube during storage and operation, thus helping to maintaining the vacuum. The barium layer also provides a clear visual indicator of the vacuum status. The silver colored barium layer will turn white if the vacuum is ever lost. This makes it easy to determine whether or not a tube is in good condition. See picture below.



The Getter is located at the bottom of the evacuated tube.



Left Tube = Vacuum Present  
Right Tube = Faulty

Evacuated tubes are aligned in parallel; the angle of mounting depends upon the latitude of your location. In a North South orientation the tubes can passively track heat from the sun all day. In an East West orientation they can track the sun all year round. The efficiency of a evacuated water heater is dependent upon a number of factors, one important one being the level of evacuated radiation (insulation) in your region.

| EVACUATED TUBE BASIC SPECIFICATIONS |                         |
|-------------------------------------|-------------------------|
| Length (nominal)                    | 2100mm                  |
| Outer tube diameter                 | 58 mm                   |
| Inner tube diameter                 | 47mm                    |
| Glass thickness                     | 1.6mm                   |
| Thermal expansion                   | 3.3x10-6 oC             |
| Material                            | Borosilicate Glass 3.3  |
| Absorptive Coating                  | Graded CU-N/CU          |
| Absorptance                         | >92% (AM1.5)            |
| Emittance                           | <8% (80°C)              |
| Vacuum                              | P<5x10 <sup>-3</sup> Pa |
| Stagnation Temperature              | >200°C                  |
| Heat Loss                           | <0.8W/ (m2oC)           |



| HOT WATER STORAGE TANK       |                                      |
|------------------------------|--------------------------------------|
| Tank Type                    | Horizontally assembled               |
| Tank Material (Inner)        | <b>GLASS LINED TANK</b>              |
| Side dish                    | Dish ended                           |
| Welding                      | Mig welding                          |
| Tank testing                 | Factory Leak tested in gravity flow  |
| Tank operating pressure      | Normal gravity flow                  |
| Outer tank gladding          | Coated Sheet                         |
| Insulation Material for tank | PUF (Poly Urethane foam)             |
| Insulation Material Density  | 48 Kgs/cu.m.                         |
| Thickness of Insulation      | 50 mm                                |
| Support stand                | Galvanised Steel with PU Coated      |
| Gaskets                      | EPDM Rubber to withstand temperature |
| Inlet & Outlet pipe ends     | 3/4" Galvanised Steel                |
| Fasteners                    | Galvanized Pipe                      |
| Heater Coil                  | Optional                             |

| SYSTEM SPECIFICATION FOR SOLAR WATER HEATER ETC MODEL |                                |
|-------------------------------------------------------|--------------------------------|
| System Capacity (Water Output)                        | 300 Litters - Per day          |
| System Temperature Output                             | 50-55 Degree Cel               |
| Type of Absorber                                      | ETC (Evacuated Tube collector) |
| No. of Tubes                                          | 24 No's                        |
| Application                                           | Bathing                        |
| Circulation                                           | Thermosyphon                   |
| Hot Water Tank Capacity                               | <b>300 Litres</b>              |



| OFFER FOR SOLAR WATER HEATER                                        |        |           |       |
|---------------------------------------------------------------------|--------|-----------|-------|
| Description                                                         | Qty    | Unit Rate | Value |
| SOLAR WATER HEATING SYSTEMS 300 LPD ETC MODEL<br>(GLASS LINED TANK) | 7 No's | 30        | 0     |
| GST @12 % ( + )                                                     |        |           | 0     |
| TOTAL AMOUNT                                                        |        |           | 0     |

## TERMS AND CONDITIONS

- **Prices** valid for 15 days from the date of the offer. After this period a re-confirmation should be taken.
- Sales tax 12% inclusive
- **Installation charges** Extra as per actual
- **Transportation charges** Extra as per actual
- **Payment Terms:** 50% as advance along with the order, balance 50% against Performa invoice,
- **Delivery:** 20-25 Days from the date of technically and commercially clear order with advance.
- **Warranty:** 2 Year for Solar water heater. Breakage of glass tube will not cover under warranty.

**(Water Hardness Less than 200 ppm)**

- **NOTE: External Hot & Cold water piping and civil work should be done by the customer Scope**

We hope the rates are competitive and we expect orders from you at the earliest and for any further clarifications please feel free to contact us.

Thanking You,

Yours faithfully,